

L1.3 Continuity

§1.8 — what you need to know

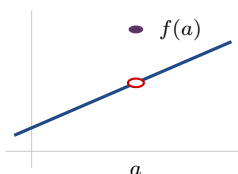
CONTINUOUS AT $x = a$

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Expanded: $\text{LHL} = f(a) = \text{RHL}$

“which means you could draw the function without picking up your pen”

Notation. LHL RHL DNE · HE @ hole · Z @ zero · VA @ · HA @



the value is there, and *wrong*

Kinds of discontinuity

removable (hole)

the value — $f(a)$ is missing — **or the match**, if $f(a)$ is there and sits at the wrong height

infinite

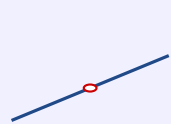
the limit — both sides run to the same infinity ($\pm\infty$), or they disagree (DNE). Either way there is no number

jump

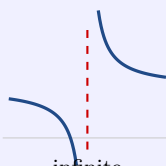
the limit — $\text{LHL} \neq \text{RHL}$, so DNE

essential (oscillatory)

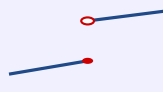
the limit — *neither* one-sided limit exists. Nothing to jump between and nothing to fill in



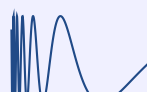
removable



infinite



jump



essential

FAMILIAR FUNCTIONS ARE CONTINUOUS EVERYWHERE ON THEIR DOMAIN

polynomial · root · trig · rational · exponential · logarithmic

So for any standard toolkit function, look for places outside the domain of the function — that is where continuity breaks. You find them by *factoring*, not by taking a limit at every point.

And the list is closed: if f and g are continuous at a , so are $f + g$ $f - g$ cf fg $\frac{f}{g}$ (if $g(a) \neq 0$) $f \circ g$. **Anything you build out of familiar functions with those six is continuous everywhere on its own domain too.**

INTERVALS OF CONTINUITY

At the edge of a domain the two-sided test **cannot be run** — there is nothing on the other side. Demanding it would call every root function broken at its own edge.

$$\text{right-continuous at } a : \lim_{x \rightarrow a^+} f(x) = f(a)$$

Mirror it for a left endpoint. $\sqrt{x-1}$ is continuous on $[1, 2]$.

This is also what decides a bracket: an interval closes with $[]$ at an endpoint the curve actually *arrives at* from inside, and with $()$ where it does not.

INTERMEDIATE VALUE THEOREM

$$f \text{ cont. on } [a, b], N \text{ between } f(a) \text{ and } f(b) \Rightarrow \exists c \in (a, b), f(c) = N$$

“it **MUST TRACE THRU**”

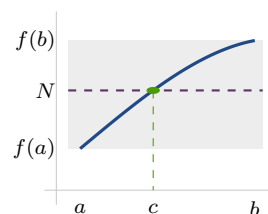
Draw a continuous function at height $f(a)$ when $x = a$ and $f(b)$ when $x = b$. To get from one to the other it has to **sweep through every height in between**.

The move — to show two things must be equal:

$$f(x) = \text{thing1}(x) - \text{thing2}(x)$$

then show f has a **root**. A root of the difference is a point where the two agree.

It gives **existence only**. Never where. Never how many. And “between” is order-free — $f(b)$ may sit below $f(a)$.



every height in the band gets hit — **at least once**

WATCH OUT

✗ ~~a corner is a discontinuity~~ → continuity is not smoothness — $|2x - 1|$ is continuous at $\frac{1}{2}$

✗ ~~removable means $f(a)$ is missing~~ → it can be there and simply *wrong* — the limit is the honest answer

✗ ~~three kinds is a theorem~~ → it is a list of the common ones — $\sin(1/x)$ at 0 is the fourth

✗ ~~you can substitute anywhere~~ → only on the domain — so look outside the domain, that is where it breaks

✗ ~~the theorem hands you the c~~ → existence only; never where, never how many

✗ ~~IVT needs $f(a) \leq N \leq f(b)$~~ → *between*, either way round — $f(b)$ may be the smaller one